

Systemic and concrete methods and tools to address environmental complexity and rebound effects within a design or decision-making process

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Keywords

sustainability, socio-technical systems,

HCI, ICT, design, decision-making,

complexity, systems thinking, systemic design, systems dynamics,

indirect effects, rebound effect, socio-environmental impacts

design methods, design cards, modelling, prospective scenarios

Laetitia Bornes

Academic background

2009-2012

Master of Engineering
ENTPE, Lyon, France

Bachelor of Architecture
ENSAL, Lyon, France

2012-2014

Master of Architecture
ENSAPVS, Paris, France

2016-2017

Master of Digital Design
Gobelins, Paris, France

2021-2024

PhD in Computer Science
ENAC & ISAE-SUPAERO

+ 3 months at Lancaster, UK
(Socio-Digital Sustainability)

2025

Independent researcher
Laetitia Bornes Design



2015

Architect
Architectes de
l'urgence,
Amiens, France

2016-2020

UX/UI Designer
Ingenuity, Toulouse, France

2018

Akiani, Bordeaux, France

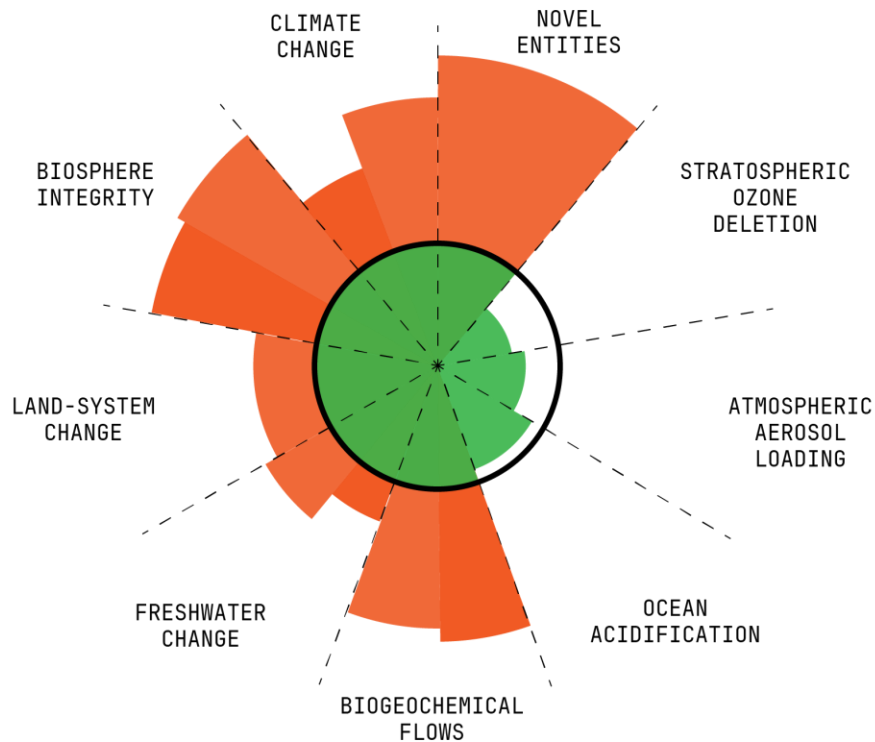
2020-2021

UX/UI Designer
Laetitia Bornes Design,
Toulouse, France

Work experience

Context & motivation

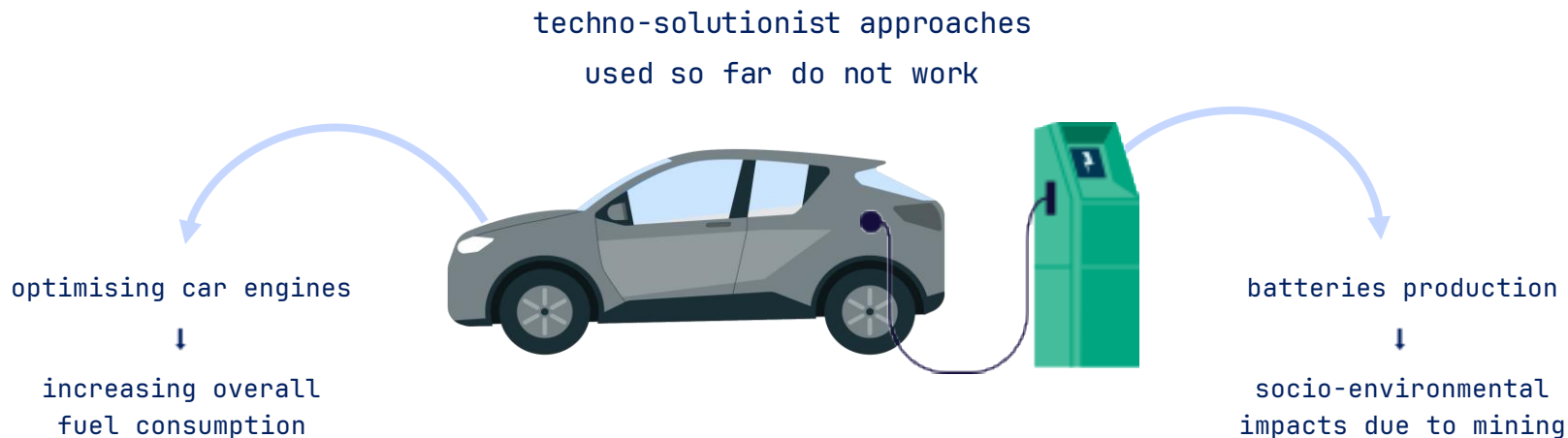
The planetary situation is alarming



The planetary boundaries framework.

Inspired from: Azote for Stockholm Resilience Centre, Stockholm University. Based on Richardson et al. 2023.

Urgent action is needed, but it's far from simple...



Faced with this urgent and complex situation, what can digital designers do?

Sustainable HCI*, ICT* and SE*

‘sustainability and unsustainability are conditions of a “system as a whole”’

[Knowles et al, 2018]

- imagining new ways of living
- interactions between sectors
- different time scales
- all socio-environmental impacts

*‘it has often fallen short on practical advice and on suitable techniques that are **concrete enough to be actionable**’*

[Raghavan & Pargman, 2017] quoted in [Bremer et al, 2022]

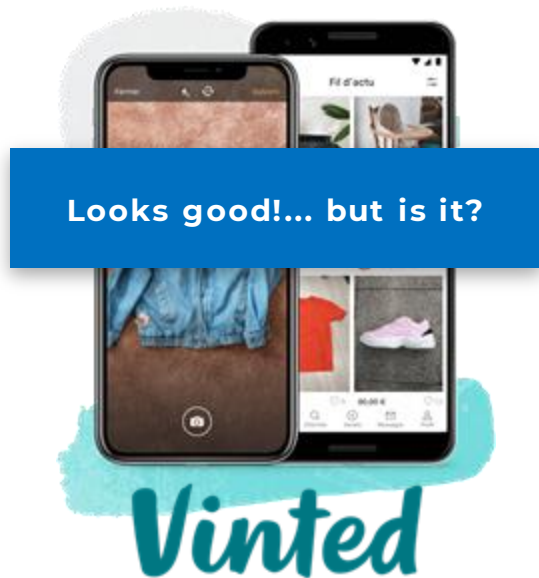
Could designers take a practical approach to such a huge problem without being reductive?

*HCI: Human-Computer Interaction

*ICT: Information and Communication Technology

*SE: Systems Engineering

Vinted | Second hand resale platform



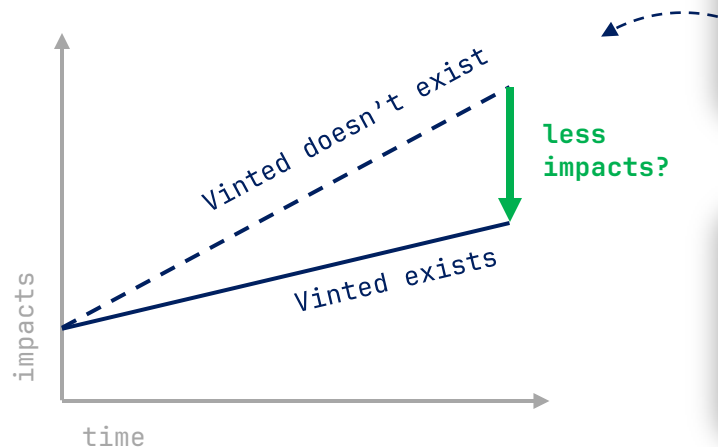
Vaayu study on Vinted avoided emissions



Avoided emissions = **453 ktCO₂eq**
(**1.8kgCO₂eq** per purchase)

Vaayu, "Vinted Climate Change Impact Report," Vaayu, 2021

What is behind this calculation?



Consequential approach

BUYERS

If Vinted didn't exist, you would have bought this item...

- (a) brand new
- (b) second hand (elsewhere)
- (c) no, I was just browsing...

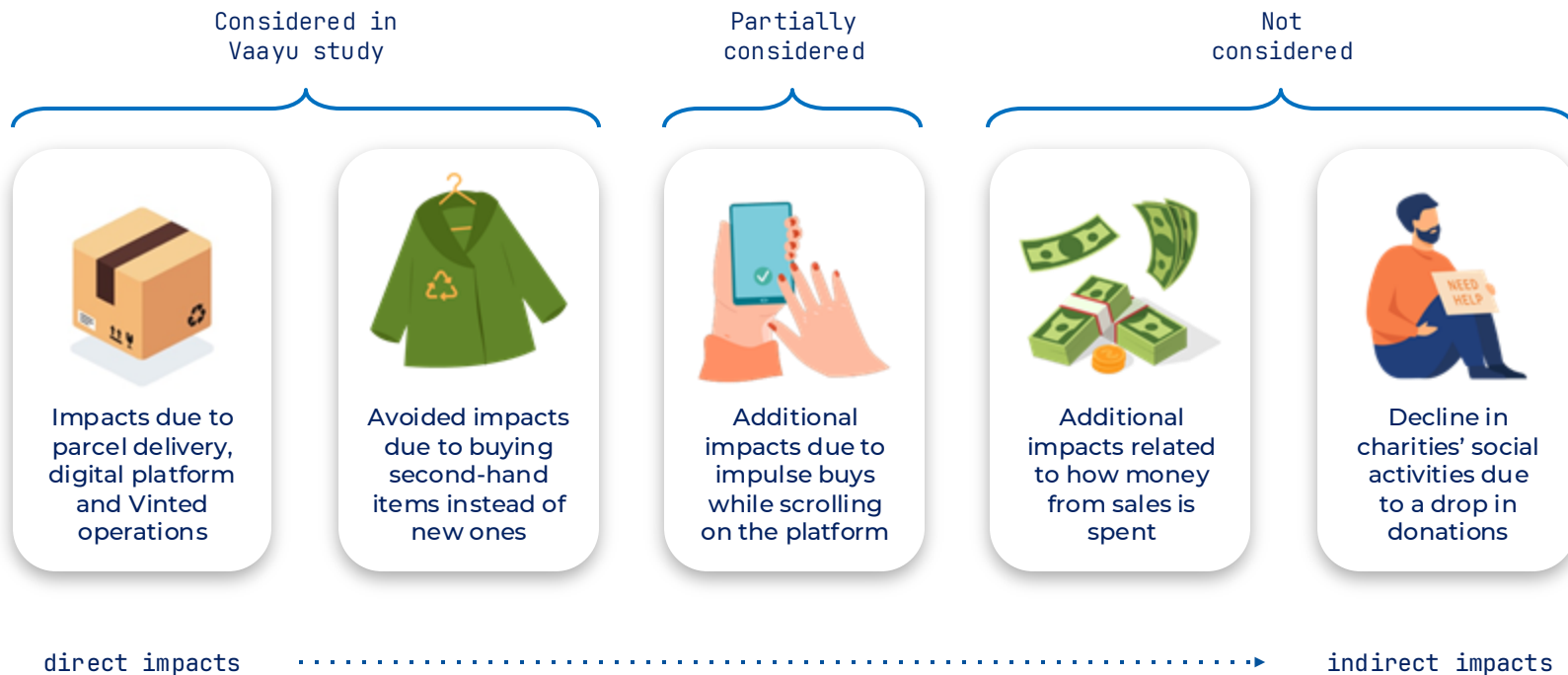
SELLERS

If Vinted didn't exist, you would have...

- (a) sold this item elsewhere
- (b) given this item to charities
- (c) thrown it...

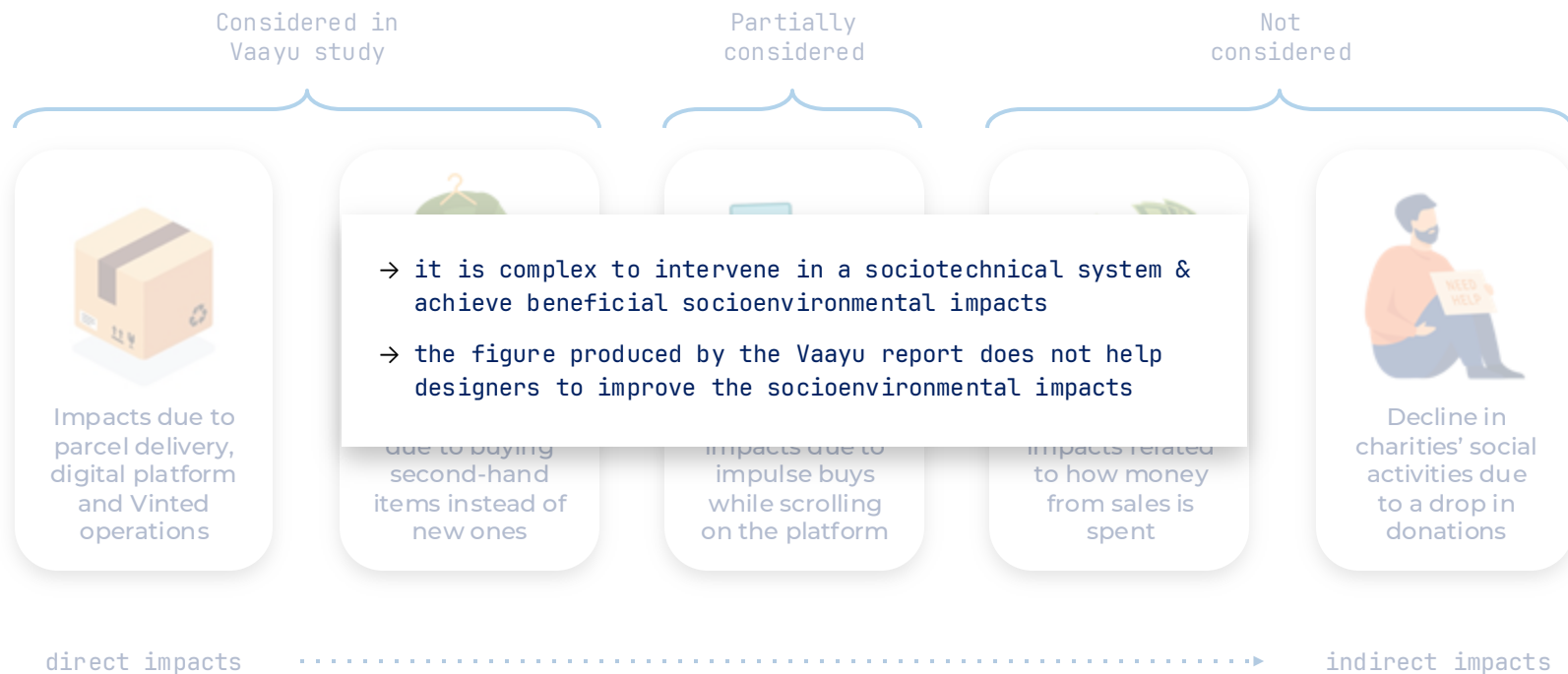
Consequential LCA based on a 350,000 Vinted users survey
Vaayu, "Vinted Climate Change Impact Report," Vaayu, 2021.

Some direct and indirect effects of Vinted



E. Juge, "La fabrique des conso-marchands: une approche par les dispositifs sociotechniques dans le contexte de la consommation collaborative," Ph.D. dissertation, Univ. of Lille, 2018.
Vaayu, "Vinted Climate Change Impact Report," Vaayu, 2021.

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Research question & Approach

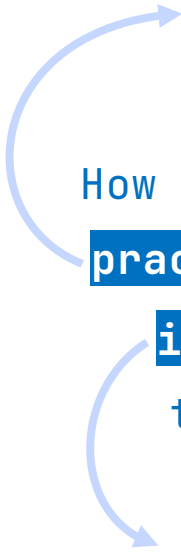
Main research question

How might designers and decision-makers be empowered with **practical methods and tools** to thoughtfully consider the **indirect and rebound effects** of their interventions throughout the design and decision-making process?

Research approach

“Systemic modelling methodology”

a methodology to help designers tackle rebound effects

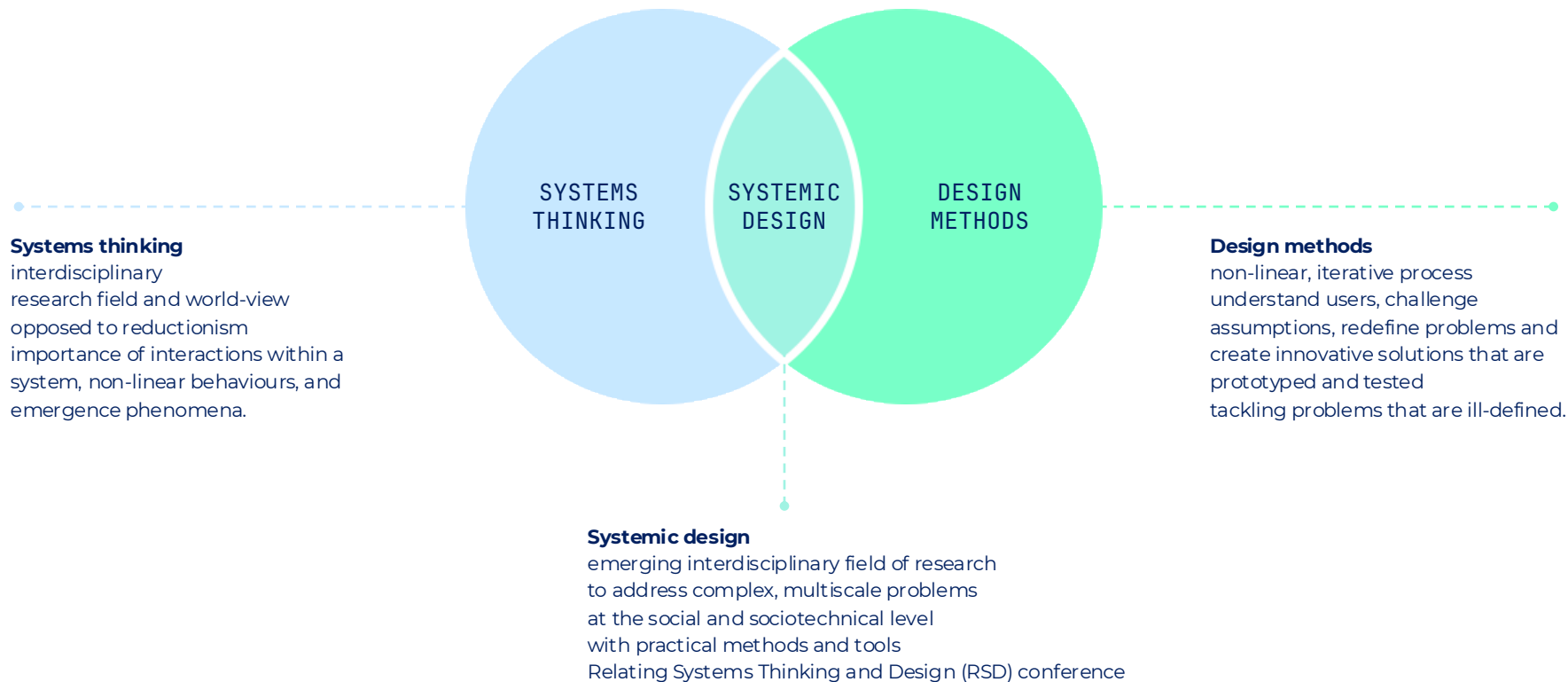


How might designers and decision-makers be empowered with **practical methods and tools** to thoughtfully consider the **indirect and rebound effects** of their interventions throughout the design and decision-making process?

Exploratory approach

constructivist-interpretive, based on *action-research*, *research through design*, and *case study* methodology

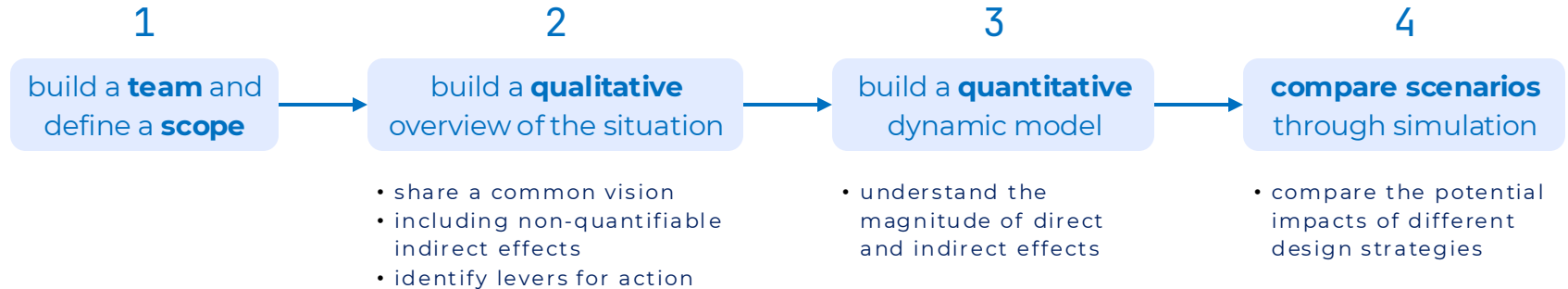
Inspired by the consequential approach & Systemic design



Systemic modelling methodology

Steps of the systemic modelling methodology

- inspired from Group Model Building -



1. Requisite variety



“getting the whole system in the room”

[Jones, 2014]

build a **team** and
define a **scope**

build a **qualitative**
overview of the situation

build a **quantitative**
dynamic model

compare scenarios
through simulation

e.g. designers, decision-
makers, academics,
policy-makers, and
citizens

could help to build a more complete
picture and reduce the bias/risk of misuse

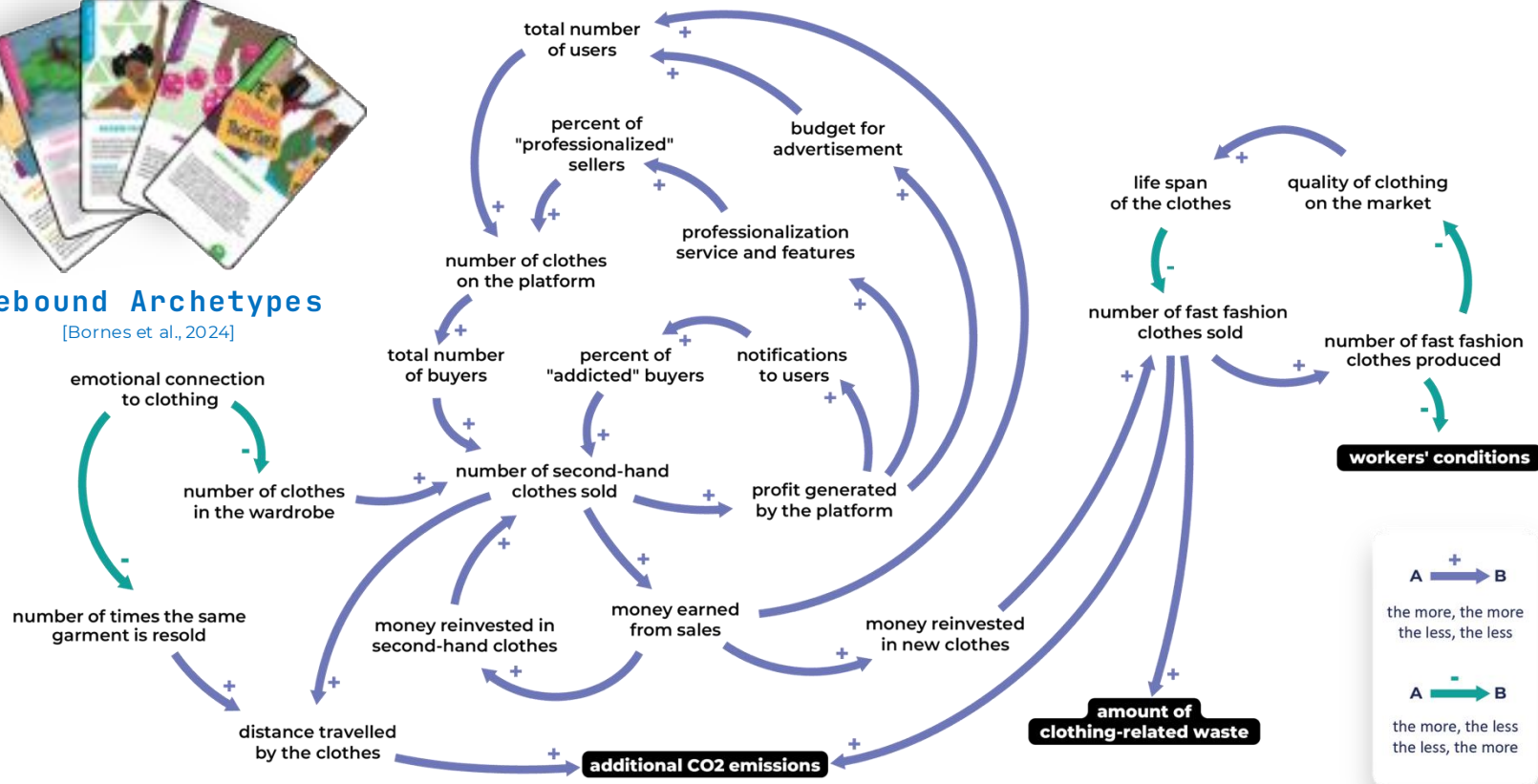
diverse interests
represented

2. Qualitative overview - Causal loop diagram

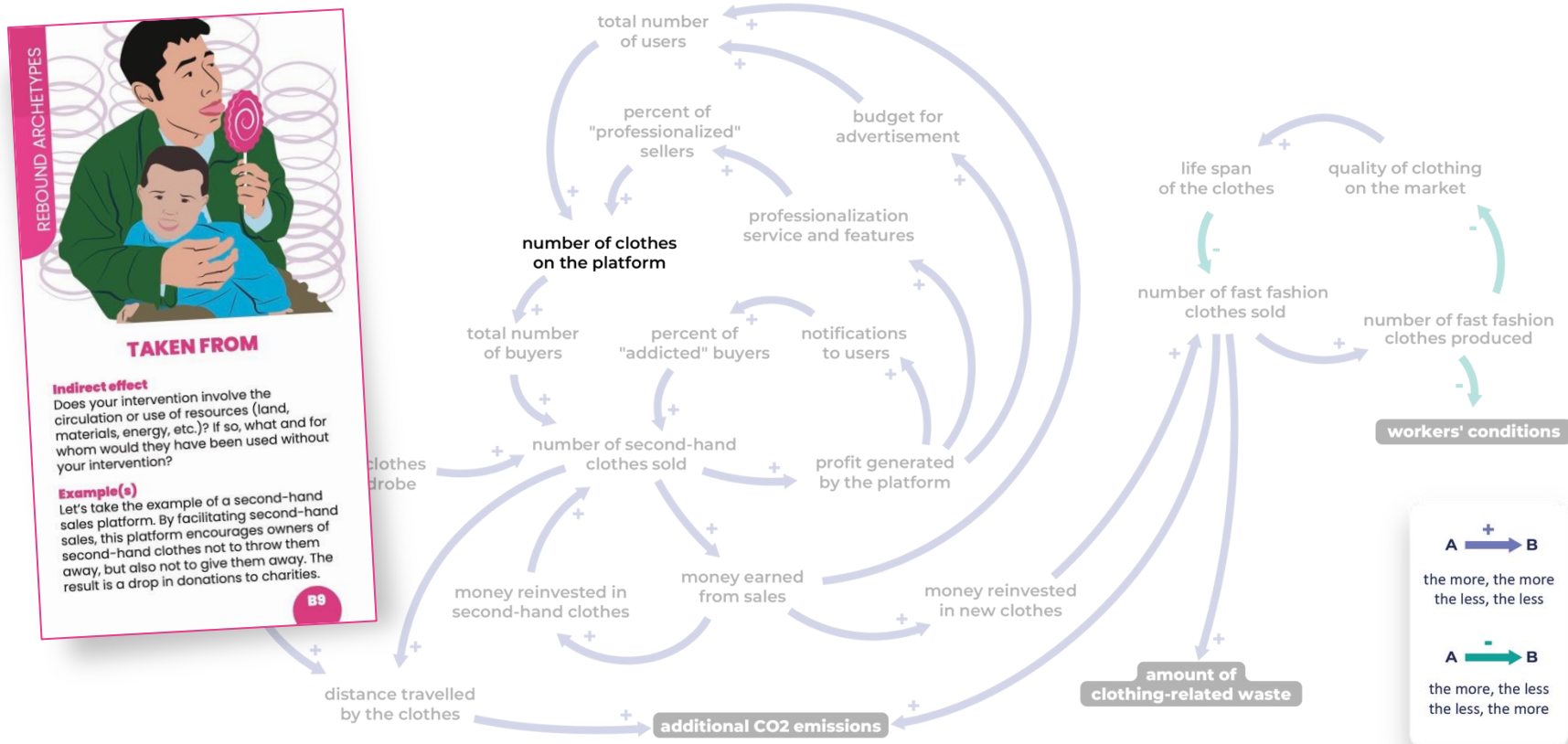


Rebound Archetypes

[Bornes et al., 2024]

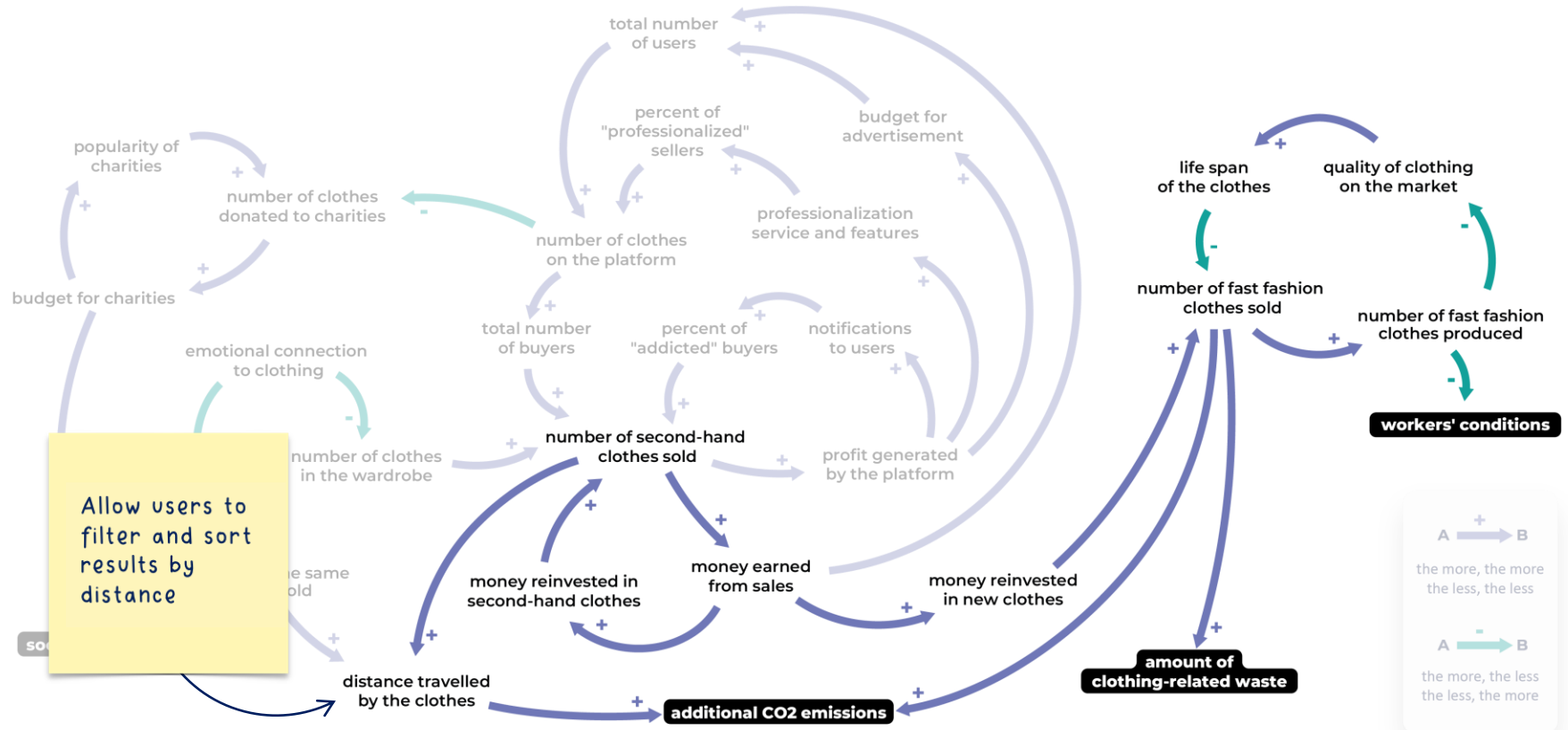


2. Qualitative overview - Causal loop diagram

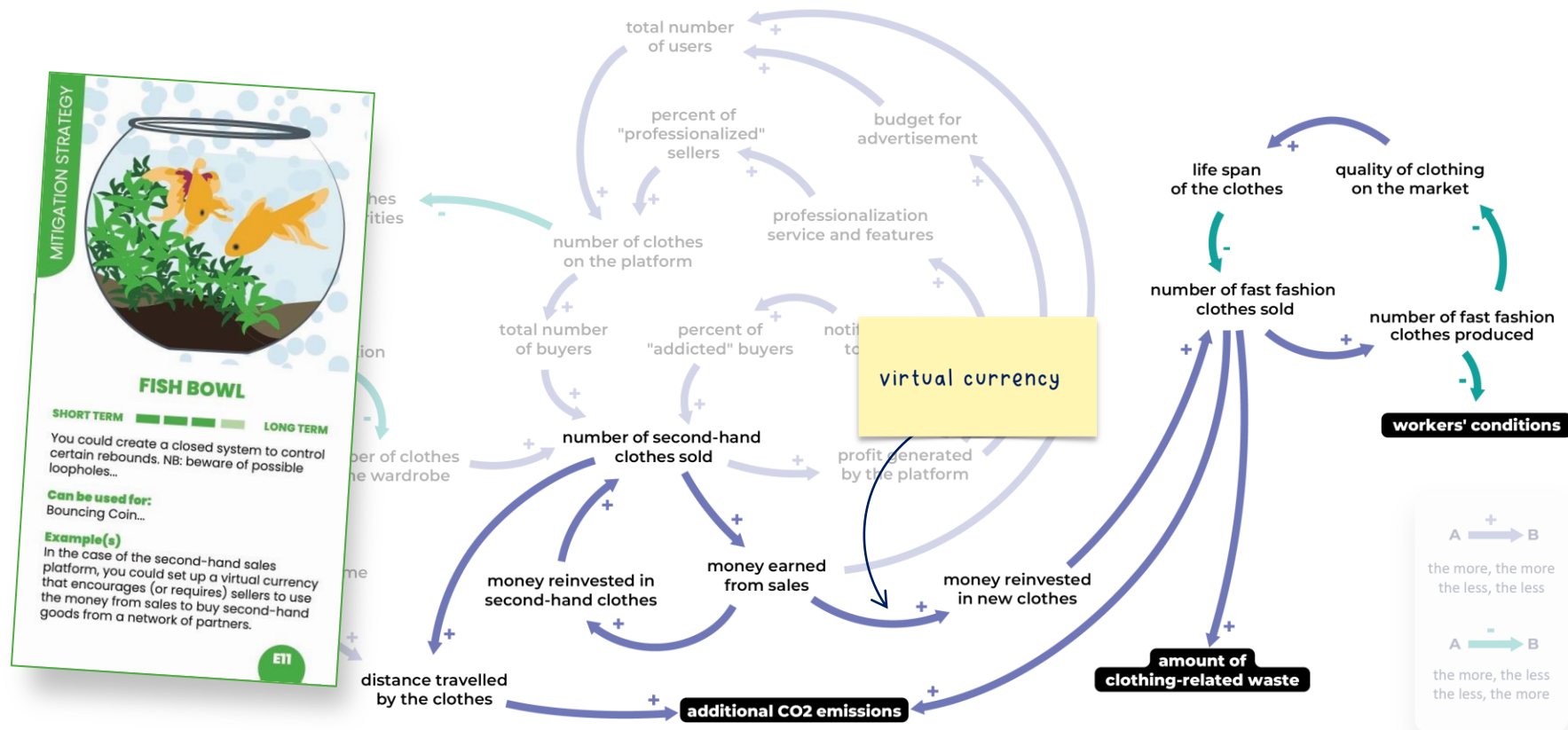




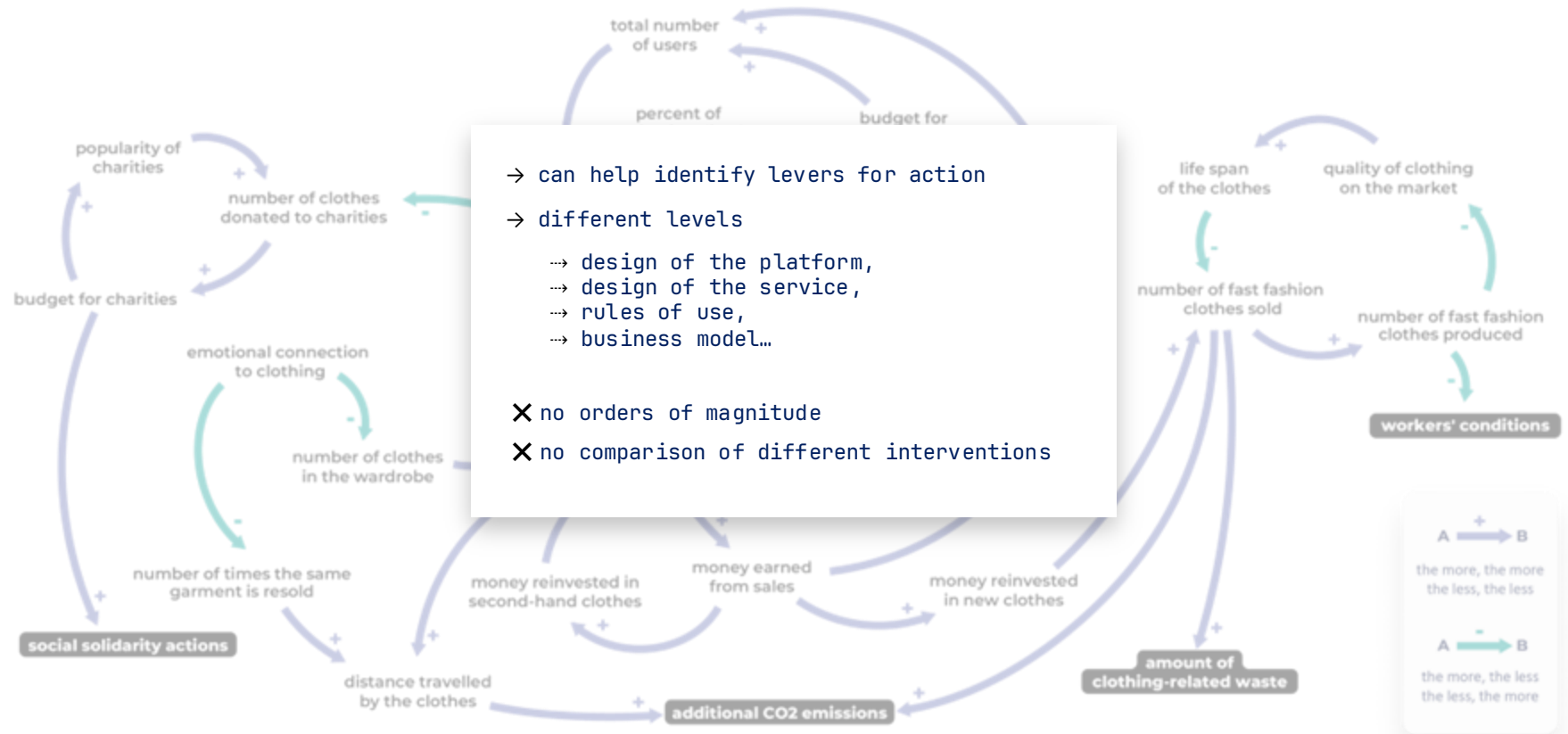
2. Qualitative overview - Causal loop diagram



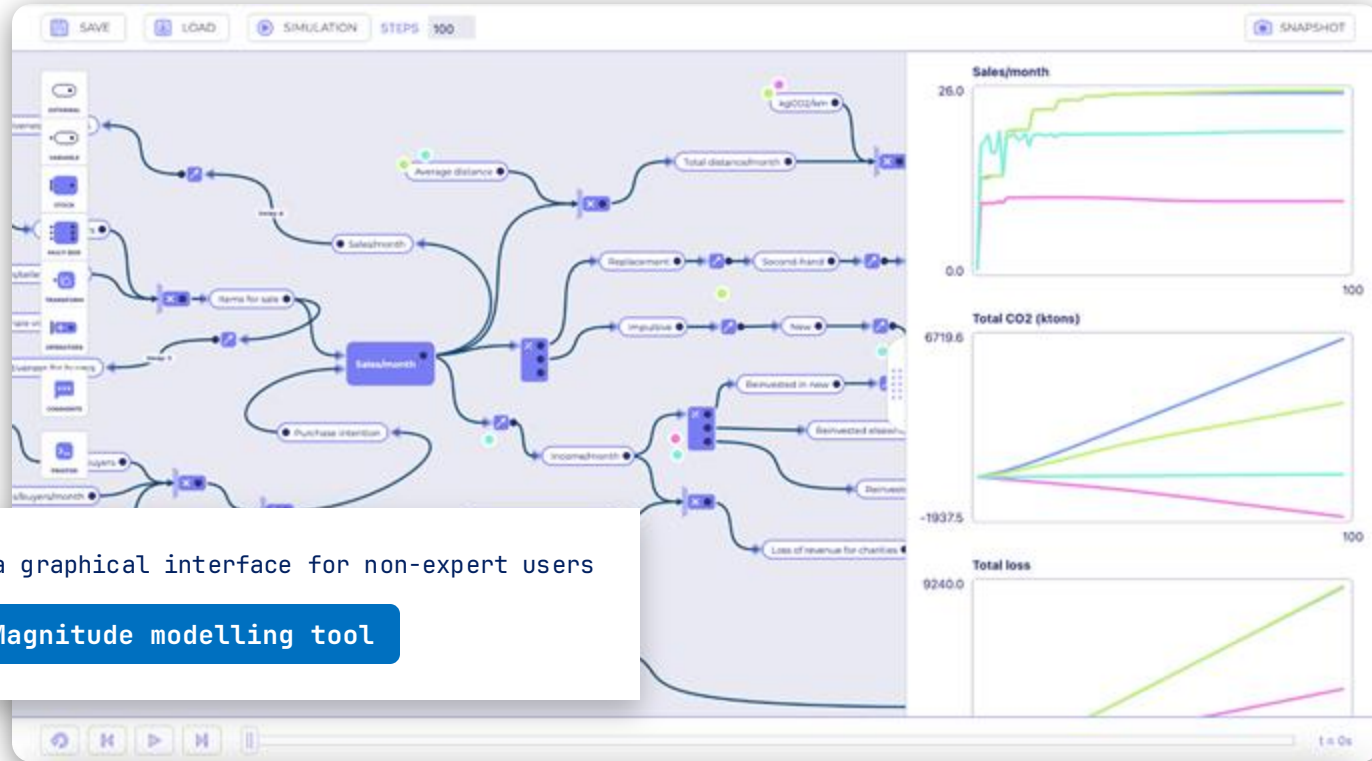
2. Qualitative overview - Causal loop diagram



2. Qualitative overview - Causal loop diagram



3. Quantitative dynamic model – Magnitude modelling tool



Magnitude modelling tool

<https://lii.enac.fr/projects/magnitude/> (interaction with smala (<http://smala.io>) & calculations in C++)



SAVE



OPEN



SIMULATION

STEPS

100



SNAPSHOT

EXTERNAL

VARIABLE

STOCK

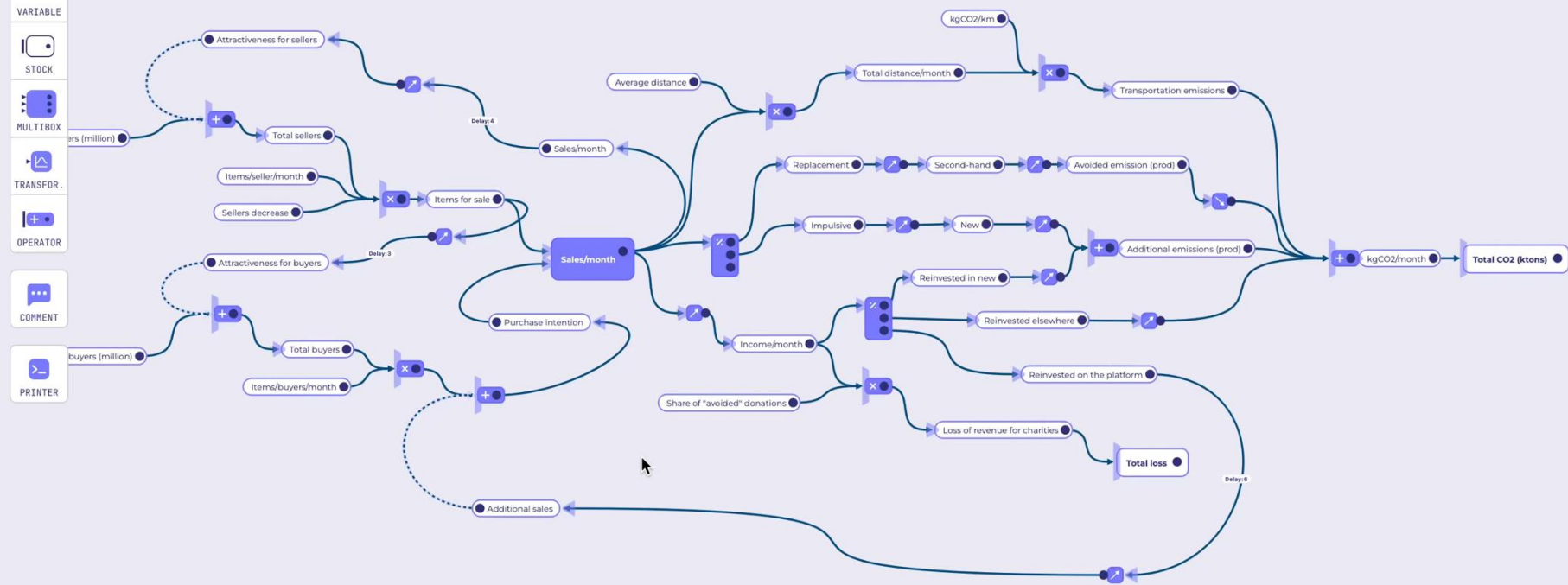
MULTIBOX

TRANSFOR.

OPERATOR

COMMENT

PRINTER



t = 0s



Good practices

identify
potential
rebound effects



Rebound Archetypes

identify
strategies to
tackle them

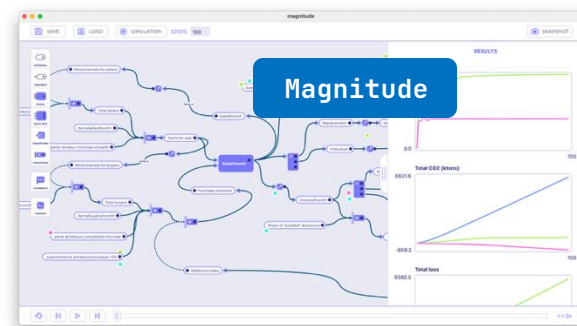
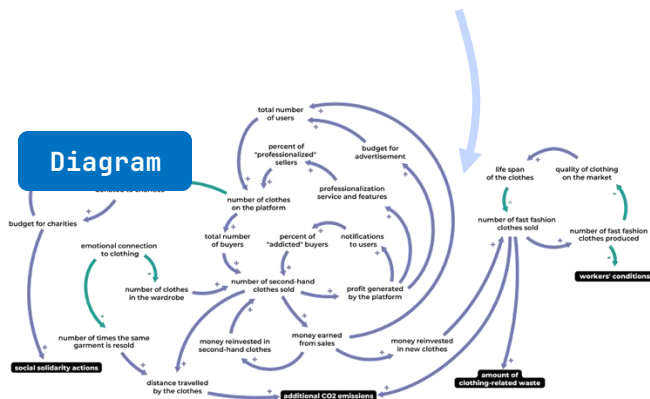
- guide stakeholders throughout the process
- can be used on their own to run quick workshops

build a **team** and
define a **scope**

build a **qualitative**
overview of the situation

build a **quantitative**
dynamic model

compare scenarios
through simulation



Cards to think through the potential indirect effects of a case study



Actants



ACTANTS

DIRECT USERS, BUYERS, CONSUMERS, ETC...

People who are directly targeted by your intervention (e.g. users of your product or service), categorised according to their relationship to it.

Example(s)
In the case of a platform selling second-hand clothes, this would be buyers and sellers. It may be useful to differentiate between professional buyers and occasional buyers.

A1

ACTANTS

PRODUCERS, PROVIDERS, MANUFACTURERS, SELLERS...

The people up and down the supply chain that directly provide elements for and/or make use of your products and services in their businesses.

Example(s)
If your product is shipped in your branded packaging, this could be the owner of the business that prints the packaging used for your products or the person that rides the bike used in the delivery service.

A2

ACTANTS

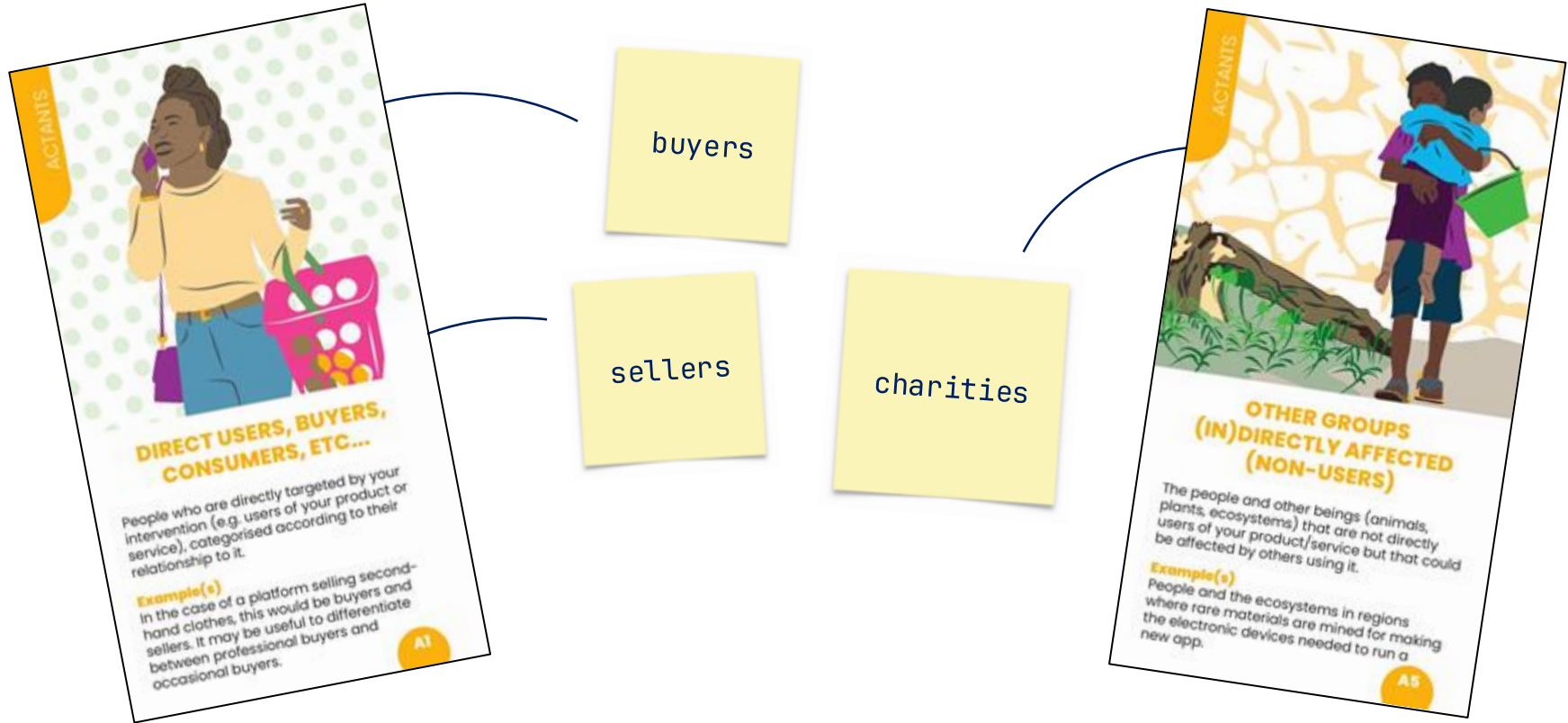
REGULATORS, GOVERNMENT, ETC...

The people working in organisations that create and monitor the rules that regulate how your product/service can be deployed and used.

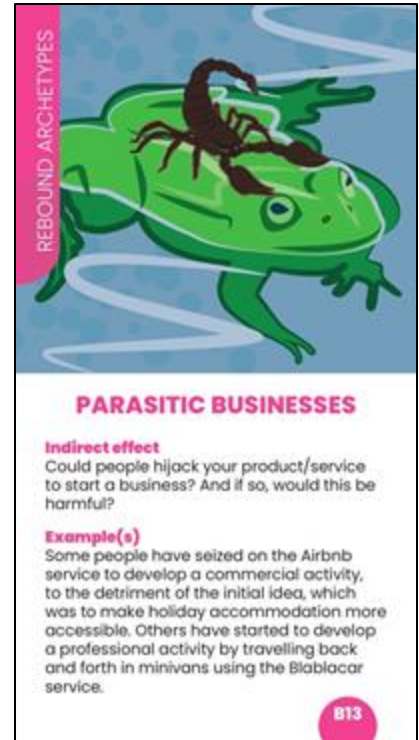
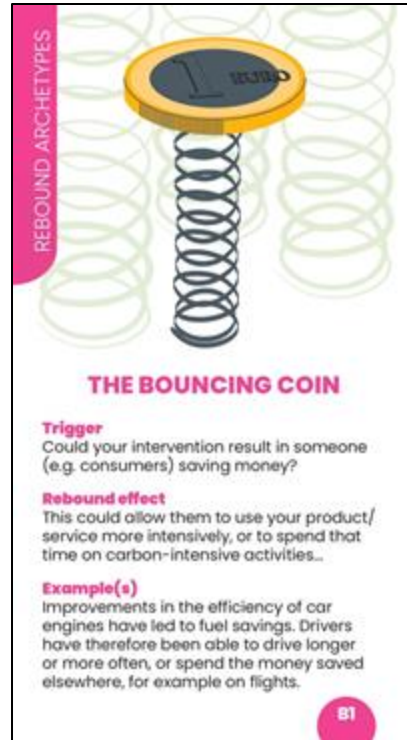
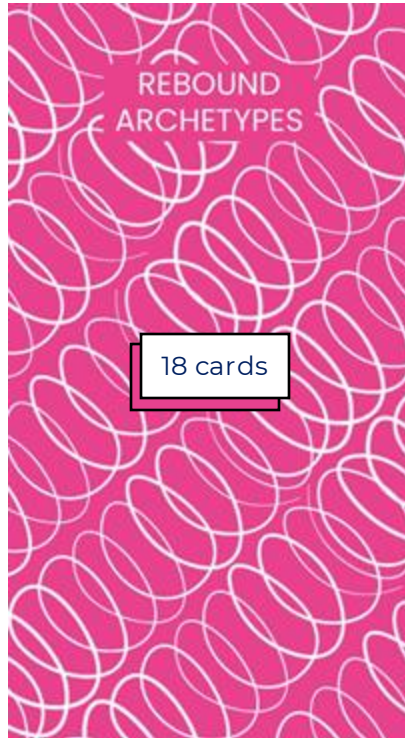
Example(s)
Policy-makers creating the laws and regulations for use of social media. Think not just locally, but also on those people working in bodies such as the European Union of the UN.

A4

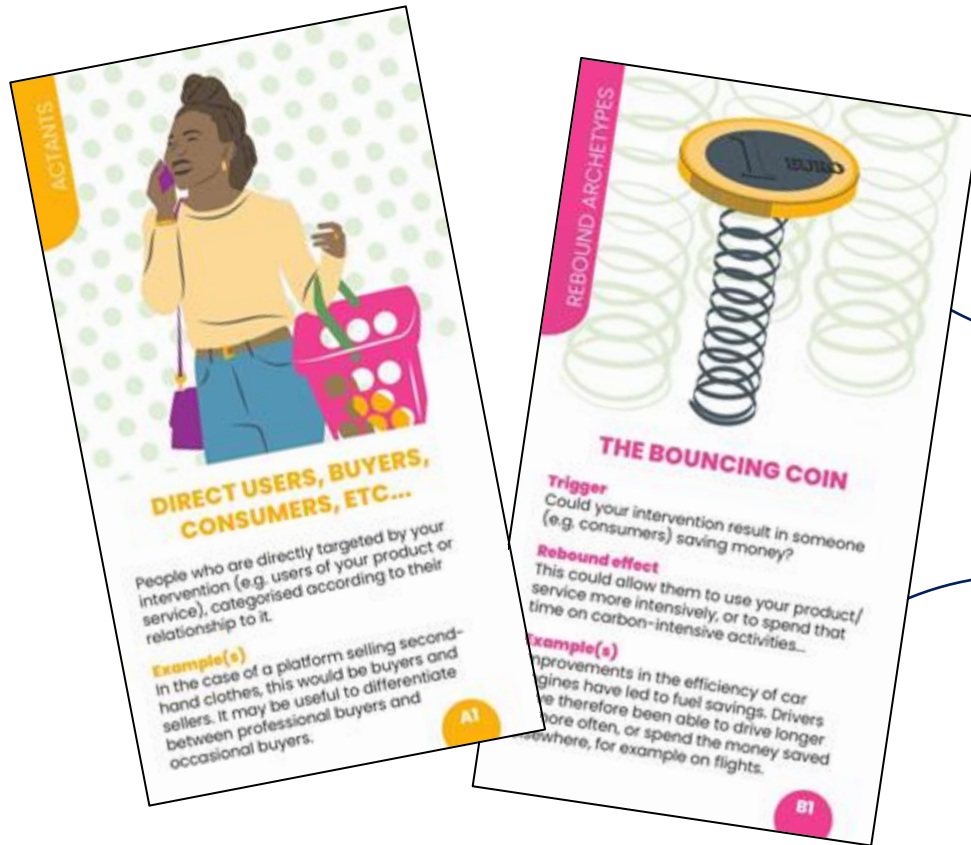
Example



Rebound Archetypes (indirect effects)



Example



buyers save money and can buy more items (B1)

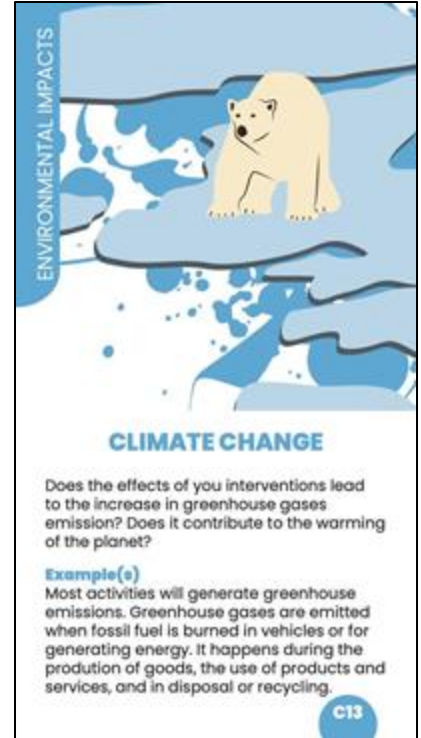
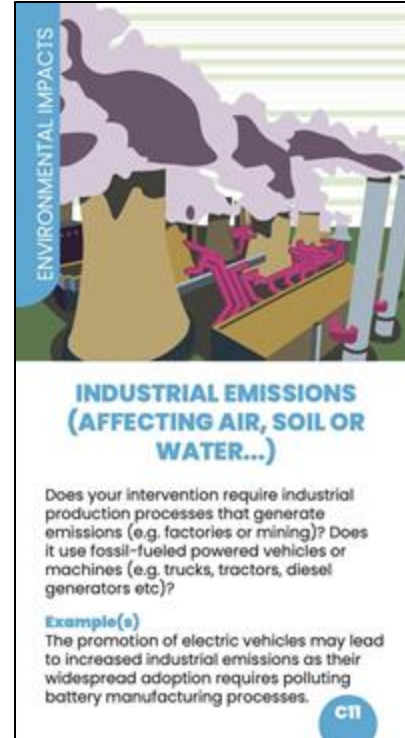
sellers earn money and can buy new fast fashion clothes (B1)

Example

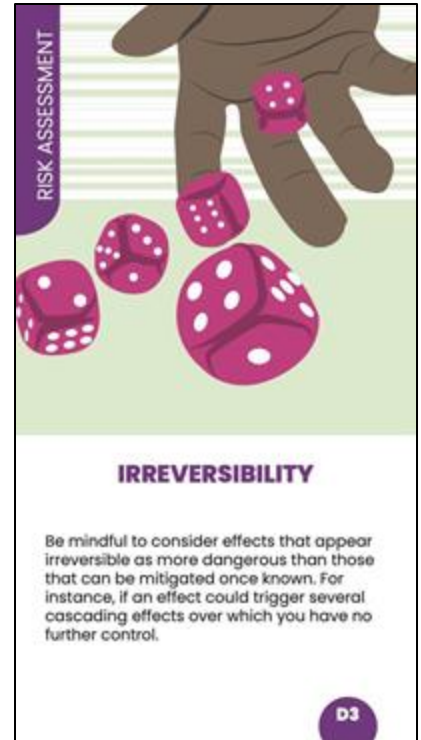
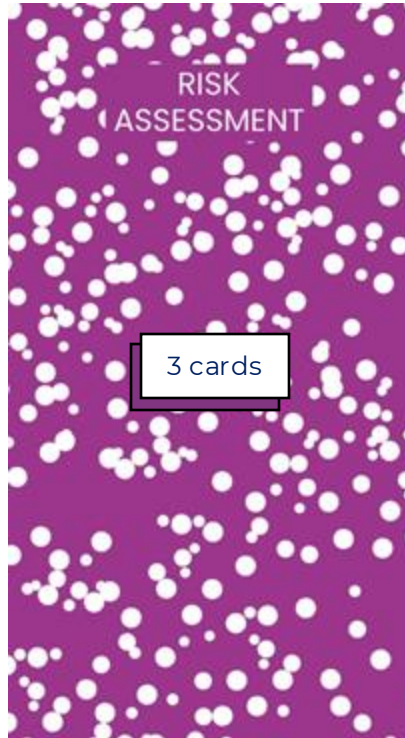


charities
receive fewer
donations,
which impact
their actions
(B9)

Socio-environmental impacts

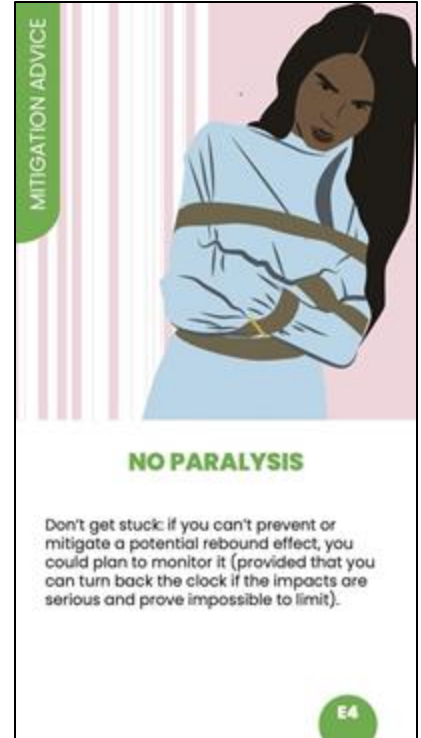
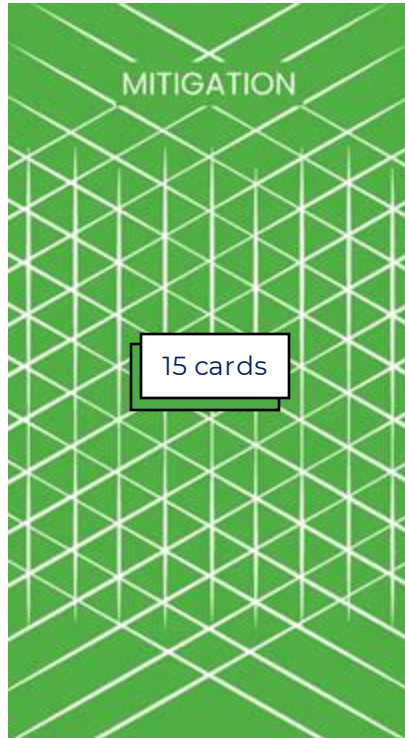


Risk assessment prompts





Decision making



Mitigation strategies



MITIGATION STRATEGY

DONATION

SPEND THAT MONEY!

SHORT TERM **LONG TERM**

You may want to encourage people to spend their money in a way that has little impact on the environment.

Can be used for:
Bouncing Coin...

Example(s)
In the case of the second-hand sales platform, this could involve, for example, telling users how much they have saved and suggesting that they donate it or spend it on a local (low-carbon) cultural event...

E7

MITIGATION STRATEGY

FISH BOWL

SHORT TERM **LONG TERM**

You could create a closed system to control certain rebounds. NB: beware of possible loopholes...

Can be used for:
Bouncing Coin...

Example(s)
In the case of the second-hand sales platform, you could set up a virtual currency that encourages (or requires) sellers to use the money from sales to buy second-hand goods from a network of partners.

E11

MITIGATION STRATEGY

PARADIGM SHIFTING

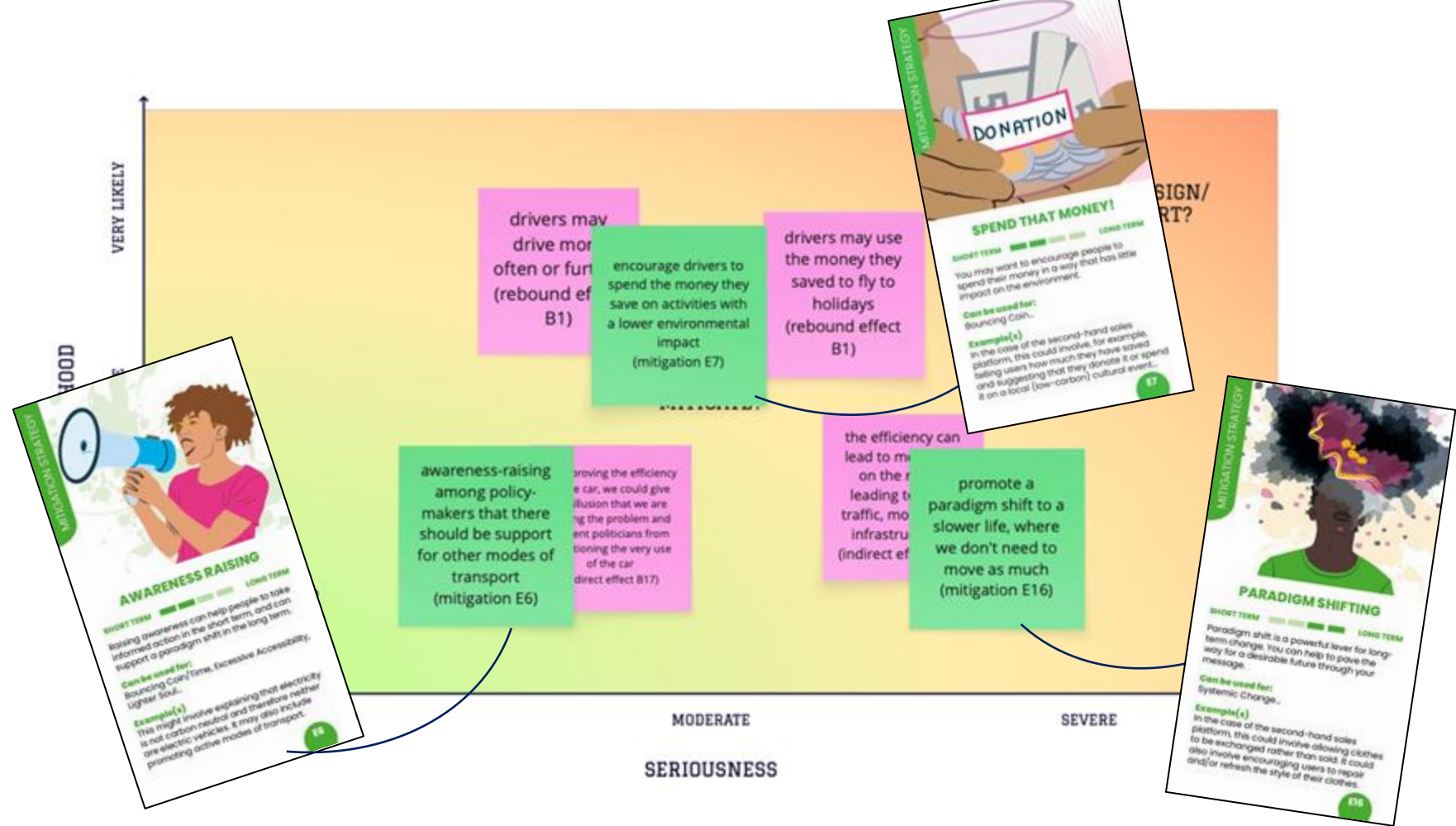
SHORT TERM **LONG TERM**

Paradigm shift is a powerful lever for long-term change. You can help to pave the way for a desirable future through your message.

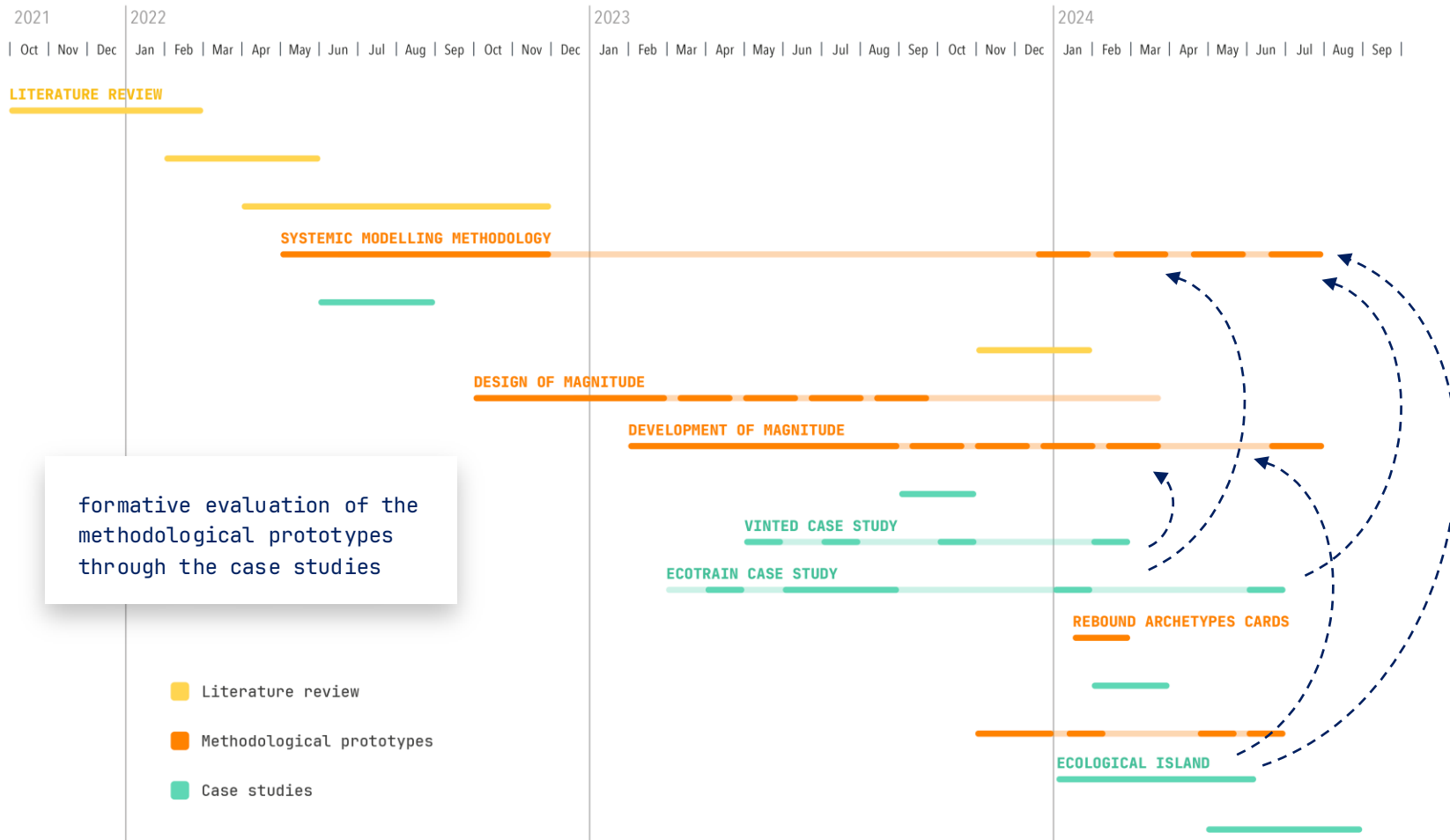
Can be used for:
Systemic Change...

Example(s)
In the case of the second-hand sales platform, this could involve allowing clothes to be exchanged rather than sold. It could also involve encouraging users to repair and/or refresh the style of their clothes.

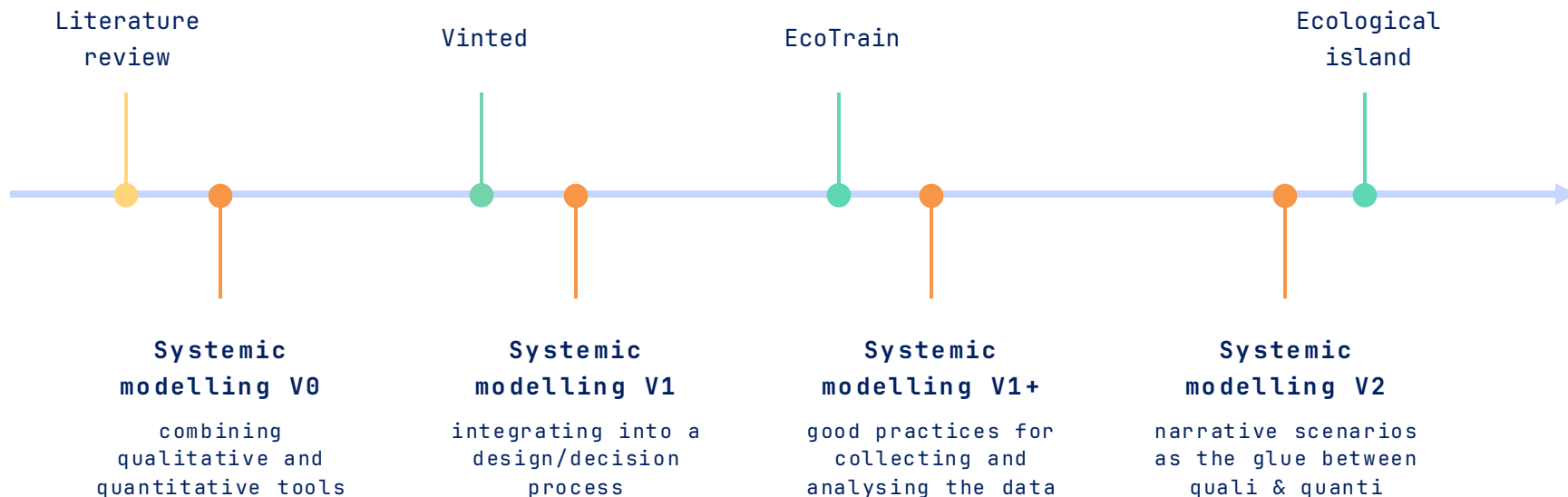
E16



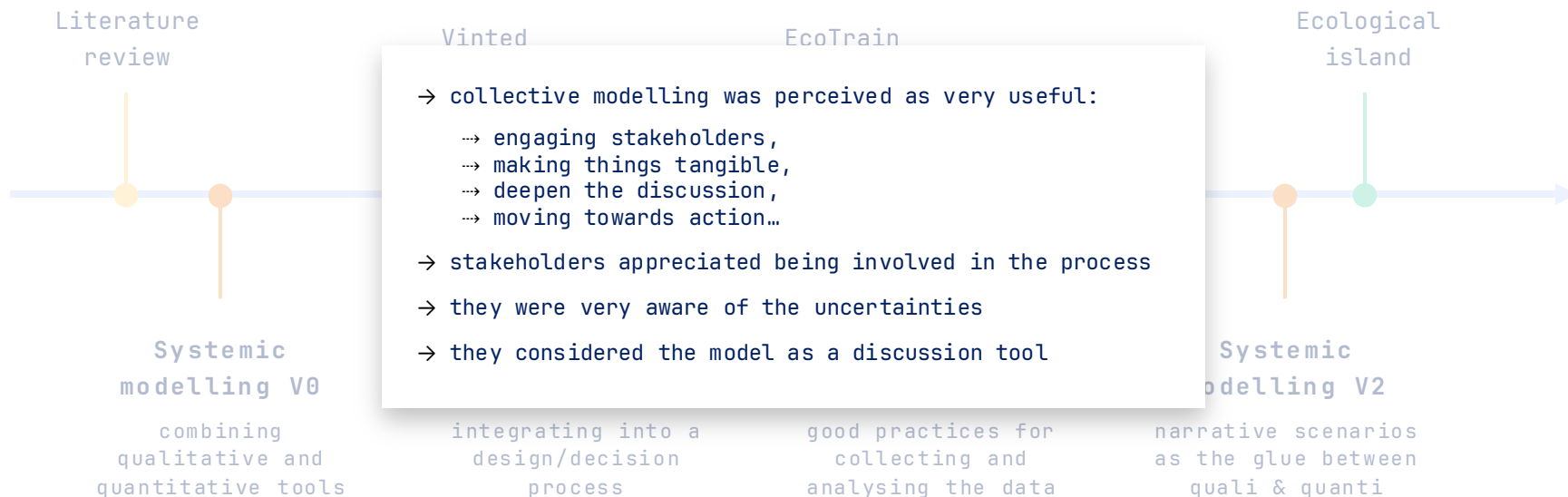
Case studies & some results



Evolution of the systemic modelling methodology



Evolution of the systemic modelling methodology



Some contributions & Perspectives

Concrete tools used by practitioners

Rebound Archetypes

Cards to print & Miro template to facilitate online workshops



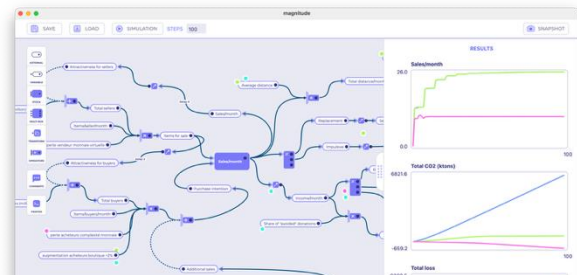
developed with
used by:



- > Designers Ethiques
- > hackathon + festival (AI impacts)
- > Master Transition (ENAC & ISAE)
- > Alt-impact (ADEME program)

Magnitude

Open access modelling tool
Models examples



developed with
used by:



- > Ctrl S (design agency)
- > DiaLog (public project)
- > business ecosystems modelling
- > people asking for the Linux version...

Challenges & recommendations for environmental decision-making

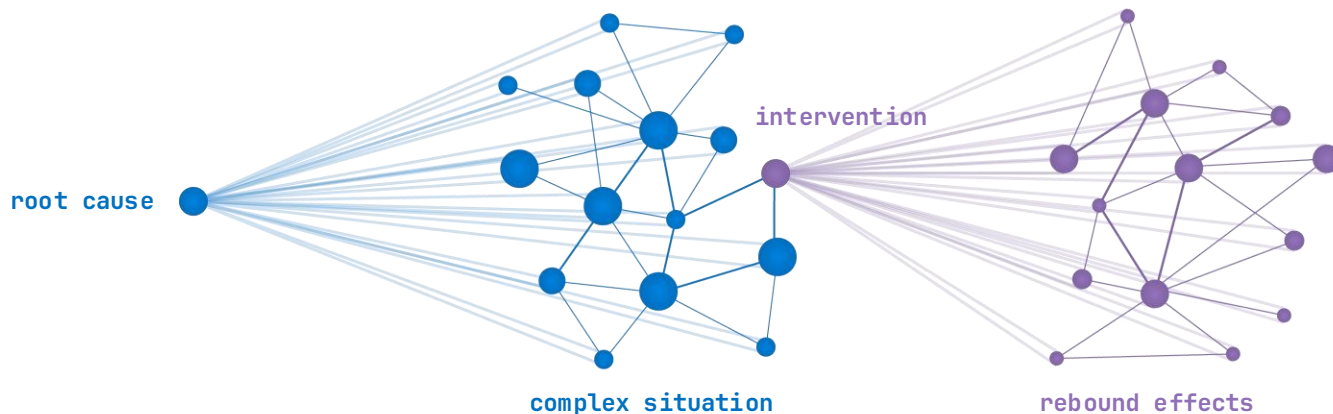
Challenges	Recommendations	Alternative tools, methods, good practices
Uncertainty	Embrace uncertainty	• Requisite variety • Causal Layered Analysis • Group Model Building
	Expose uncertainty	• Causal Loop Diagram • Systems dynamics • Model exploration
Complexity	Problem-oriented vision	• Group Model Building • Causal Loop Diagram
	Comprehension of the dynamics	• Systems dynamics modelling & simulation
Inspiration	Define an objective and a pathway	• Normative scenarios & backcasting
	Identify levers and anticipate effects	• Causal Layered Analysis • Causal Loop Diagram • Meadows' twelve leverage points • Quantitative modeling
	Support decision rather than claims	• Comparing several scenarios rather than comparing one to a hypothetical baseline
Robustness	Plurality of perspectives	• Requisite variety • Value Sensitive Design

Ekchajzer, D., Bornes, L., Combaz, J., Letondal, C., & Vingerhoeds, R. (2024, June). Decision-making under environmental complexity: the need for moving from avoided impacts of ICT solutions to systems thinking approaches. In 2024 International Conference on ICT for Sustainability (ICT4S).

Rebound effects are inherently systemic...

Rebound effects are *'the ways in which a system resists the change that you're trying to make'*.

[quote from an interview with a systemic designer]



Considering rebound effects is a systemic & concrete approach to sustainability

Some of the research avenues

- Furthering and exploring other methodological opportunities
 - Building on these findings to move towards a democratic dimension
 - Working on scenarios for multi-criteria decision-making
 - scenarios facilitate dealing with complexity and understanding future alternatives
 - prospective scenarios could be a key-tool to inform the public debate and societal choices
- How are prospective scenarios constructed (IPCC, ADEME, etc.)?
 - How are prospective scenarios used?
 - Could and should we rethink the methodology used to build these scenarios?
 - What role HCI could play in making them interactive and more transparent?



PEPR eSEMBLE

Collaborating on IPCC scenarios: Rethinking digital tools and human processes to meet current challenges

Scenarios are the ‘backbone’ of collaboration between IPCC WGs (Meinshausen et al., 2024).

→ Rethinking the collaboration on these scenarios...

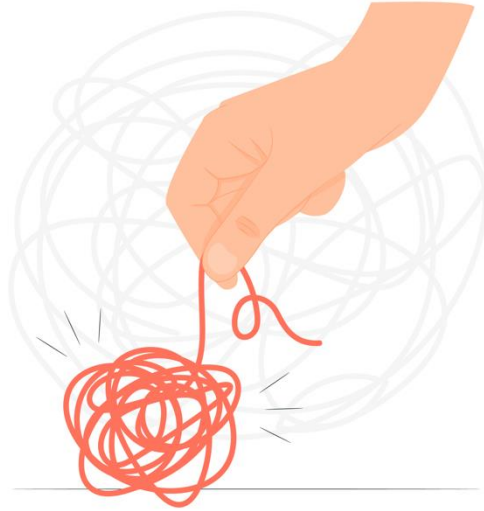
- rethinking human processes and digital tools

→ ... to meet current challenges

- volume and diversity of the literature (Hermansen et al., 2023).
- inadequacy of the length of assessment cycles with the urgency of the situation.
- political inaction / destabilisation of the IPCC by the Trump administration (Tollefson, 2025).

→ Opportunities identified during a feedback workshop on the 6th assessment cycle

- setting up a ‘community-driven database’.
- need for a digital tool but also ‘a framework and a formalised process’ (Pirani, 2024).



Collective modelling as a means to engage with a complex situation

Thank you!

Publications

Long paper **Bornes, L***, Smith M.*, Bates O., Blair G., Letondal C., and Vingerhoeds R. (2024, October). Rebound Archetypes: A Card-based Tool to Help Designers Think Through the Rebound Effects when Designing for Sustainability. In Proceedings of Relating Systems Thinking and Design (RSD13) Symposium. * Authors contributed equally

Long paper **Bornes, L**, Letondal, C., Vingerhoeds, R. (2024, July). Systemic Sustainable HCI: Integrating Collaborative Modeling into a Design Process to Address Rebound Effects. In Proceedings of the 2024 ACM Designing Interactive Systems Conference.

Long paper (best paper award) Ekchajzer, D.*, **Bornes, L***, Combaz, J.*, Letondal, C., Vingerhoeds, R. (2024, June). Decision-Making under Environmental Complexity: Shifting from Avoided Impacts of ICT Solutions to Systems Thinking Approaches. In Proceedings of the 11th International Conference on ICT for Sustainability. * Authors contributed equally

Long paper **Bornes, L**, Letondal, C., Vingerhoeds, R. (2022, October). Could Systemic Design Methods Support Sustainable Design of Interactive Systems?. Proceedings of Relating Systems Thinking and Design (RSD11) Symposium.

Short paper (journal) **Bornes, L**, Letondal, C., & Vingerhoeds, R. (2023). Understanding the Indirect Effects of Interactive Systems Within Systems of Systems. INSIGHT, 26(4), 18-21.

Short paper Letondal, C., Laplace, I., Druot, T., Bieder, C., Pauchet, S., & **Bornes, L** (2024, April). Approches alternatives pour penser et construire le futur du transport aérien : exemples d'expériences pédagogiques. In Entretiens de Toulouse.

Short paper Letondal, C., **Bornes, L**, Garcia, J., Duchevet, A., Conversy, S., Pauchet, S., & Vo, D. B. (2024, March). Un cockpit pour l'aviation du futur ? Par quel prisme de l'IHM approcher la question ?. In IHM'24-35e Conférence Internationale Francophone sur l'Interaction Humain-Machine.

Extended abstract (CHI Doctoral consortium) **Bornes, L** (2023, April). A Methodology and a Tool to Support the Sustainable Design of Interactive Systems: Adapting systemic design to ols to model complexity in interaction design. In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (pp.1-5).

Workshop (animation) **Bornes, L**, Letondal, C., & Vingerhoeds, R. (2023, October). Using a Quali-Quantitative Modelling Tool to Explore Scenarios for More-Than-Sustainable Design. In Proceedings of Relating Systems Thinking and Design (RSD12) Symposium.

Poster (best poster award) Best poster award at ISAE PhD day. (2023, June).

Poster (best poster award) Best poster award at a AFIS (Systems engineering) conference. (2022, December).

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